```
(II) LoadModule: "intel"
(II) Loading /usr/lib/xorg/modules/drivers/intel_drv.so
```

EXTRA

Proprietary drivers

Some video card makers (most notably NVIDIA) refuse to publish the hardware specifications that would be required to implement good free drivers. They do, however, provide proprietary drivers that allow using their hardware. This policy is nefarious, because even when the provided driver exists, it is usually not as polished as it should be; more importantly, it does not necessarily follow the X.org updates, which may prevent the latest available driver from loading correctly (or at all). We cannot condone this behavior, and we recommend you avoid these makers and favor more cooperative manufacturers.

If you still end up with such a card, you will find the required packages in the *non-free* section: *nvidia-driver* for NVIDIA cards. It requires a matching kernel module. Building the module can be automated by installing the package *nvidia-kernel-dkms* (for NVIDIA).

The "nouveau" project aims to develop a free software driver for NVIDIA cards and is the default driver that you get for those cards in Debian. In general, its feature set and performance do not match the proprietary driver. In the developers' defense, we should mention that the required information can only be gathered by reverse engineering, which makes things difficult. The free drivers for ATI video cards, called "radeon" and "amdgpu", are much better in that regard although it often requires non-free firmware from the *firmware-amd-graphics* package.

13.2. Customizing the Graphical Interface

13.2.1. Choosing a Display Manager

The graphical interface only provides display space. Running the X server by itself only leads to an empty screen, which is why most installations use a *display manager* to display a user authentication screen and start the graphical desktop once the user has authenticated. The three most popular display managers in current use are *gdm3* (*GNOME Display Manager*), *sddm* (suggested for KDE Plasma) and *lightdm* (*Light Display Manager*). Since the Falcot Corp administrators have opted to use the GNOME desktop environment, they logically picked `gdm3` as a display manager too. The `/etc/gdm3/daemon.conf` configuration file has many options (the list can be found in the `/usr/share/gdm/gdm.schemas` schema file) to control its behaviour while `/etc/gdm3/greeter.dconf-defaults` contains settings for the greeter "session" (more than just a login window, it is a limited desktop with power management and accessibility related tools). Note that some of the most useful settings for end-users can be tweaked with GNOME's control center.

13.2.2. Choosing a Window Manager

Since each graphical desktop provides its own window manager, which window manager you choose is usually influenced by which desktop you have selected. GNOME uses the `mutter` win-

dow manager, Plasma uses `kwin`, and Xfce (which we present later) has `xfwm`. The Unix philosophy always allows using one's window manager of choice, but following the recommendations allows an administrator to best take advantage of the integration efforts led by each project.

BACK TO BASICS

Window manager

The window manager displays the "decorations" around the windows belonging to the currently running applications, which includes frames and the title bar. It also allows reducing, restoring, maximizing, and hiding windows. Most window managers also provide a menu that pops up when the desktop is clicked in a specific way. This menu provides the means to close the window manager session, start new applications, and in some cases, change to another window manager (if installed).

Older computers may, however, have a hard time running heavyweight graphical desktop environments. In these cases, a lighter configuration should be used. "Light" (or small footprint) window managers include WindowMaker (in the *wmaker* package), Afterstep, fvwm, icewm, blackbox, fluxbox, or openbox. In these cases, the system should be configured so that the appropriate window manager gets precedence; the standard way is to change the `x-window-manager` alternative with the command `update-alternatives --config x-window-manager`.

DEBIAN SPECIFICITY

Alternatives

The Debian policy lists a number of standardized commands able to perform a particular action. For example, the `x-window-manager` command invokes a window manager. But Debian does not assign this command to a fixed window manager. The administrator can choose which manager it should invoke.

For each window manager, the relevant package therefore registers the appropriate command as a possible choice for `x-window-manager` along with an associated priority. Barring explicit configuration by the administrator, this priority allows picking the best installed window manager when the generic command is run.

Both the registration of commands and the explicit configuration involve the `update-alternatives` script. Choosing where a symbolic command points at is a simple matter of running `update-alternatives --config` *`symbolic-command`*. The `update-alternatives` script creates (and maintains) symbolic links in the `/etc/alternatives/` directory, which in turn references the location of the executable. As time passes, packages are installed or removed, and/or the administrator makes explicit changes to the configuration. When a package providing an alternative is removed, the alternative automatically goes to the next best choice among the remaining possible commands.

Not all symbolic commands are explicitly listed by the Debian policy; some Debian package maintainers deliberately chose to use this mechanism in less straightforward cases where it still brings interesting flexibility (examples include `x-www-browser`, `www-browser`, `cc`, `c++`, `awk`, and so on).

13.2.3. Menu Management

Modern desktop environments and many window managers provide menus listing the available applications for the user. In order to keep menus up-to-date in relation to the actual set of available applications, each package usually provides a `.desktop` file in `/usr/share/applications`. The format of those files has been standardized by FreeDesktop.org:

➨ https://standards.freedesktop.org/desktop-entry-spec/latest/

The applications menus can be further customized by administrators through system-wide configuration files as described by the "Desktop Menu Specification". End-users can also customize the menus with graphical tools such as *kmenuedit* (in Plasma), *alacarte* (in GNOME) or *menulibre*.

➨ https://standards.freedesktop.org/menu-spec/latest/

HISTORY

The Debian menu system

Historically — way before the FreeDesktop.org standards emerged — Debian had invented its own menu system where each package provided a generic description of the desired menu entries in `/usr/share/menu/`. This tool is still available in Debian (in the *menu* package) but it is only marginally useful since package maintainers are encouraged to rely on `.desktop` files instead.

13.3. Graphical Desktops

The free graphical desktop field is dominated by two large software collections: GNOME and Plasma by KDE. Both of them are very popular. This is rather a rare instance in the free software world; the Apache web server, for instance, has very few peers.

This diversity is rooted in history. Plasma (initially only KDE, which is now the name of the community) was the first graphical desktop project, but it chose the Qt graphical toolkit and that choice wasn't acceptable for a large number of developers. Qt was not free software at the time, and GNOME was started based on the GTK+ toolkit. Qt has since become free software, but the projects still evolved in parallel.

The GNOME and KDE communities still work together: under the FreeDesktop.org umbrella, the projects collaborated in defining standards for interoperability across applications.

Choosing "the best" graphical desktop is a sensitive topic which we prefer to steer clear of. We will merely describe the many possibilities and give a few pointers for further thoughts. The best choice will be the one you make after some experimentation.

13.3.1. GNOME

Debian *Buster* includes GNOME version 3.30, which can be installed by a simple `apt install gnome` (it can also be installed by selecting the "Debian desktop environment" task).

GNOME is noteworthy for its efforts in usability and accessibility. Design professionals have been involved in writing its standards and recommendations, which has helped developers to create satisfying graphical user interfaces. The project also gets encouragement from the big players of computing, such as Intel, IBM, Oracle, Novell, and of course, various Linux distributions. Finally, many programming languages can be used in developing applications interfacing to GNOME.

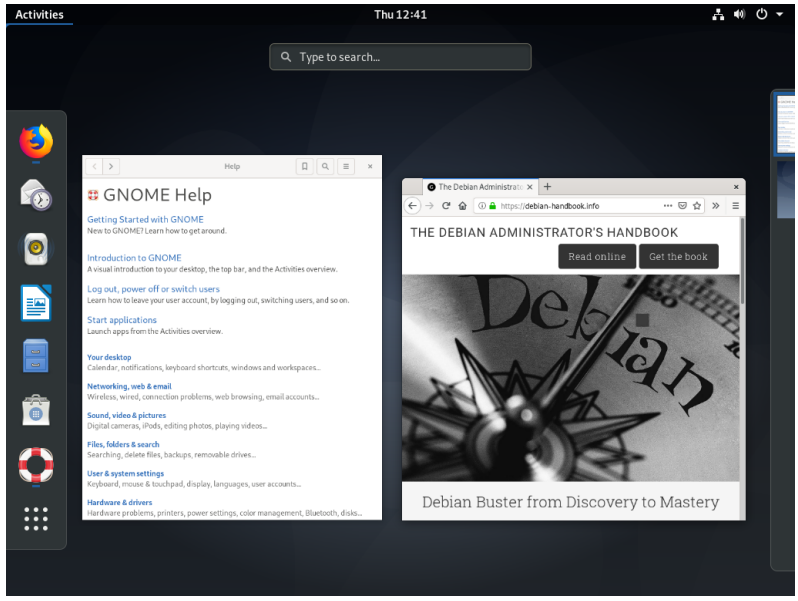


Figure 13.1 *The GNOME desktop*

For administrators, GNOME seems to be better prepared for massive deployments. Application configuration is handled through the GSettings interface and stores its data in the DConf database. The configuration settings can thus be queried and edited with the `gsettings`, and `dconf` command-line tools, or by the `dconf-editor` graphical user interfaces. The administrator can therefore change users' configuration with a simple script. The GNOME website provides information to guide administrators who manage GNOME workstations:

➨ `https://help.gnome.org/admin/`

13.3.2. KDE and Plasma

Debian *Buster* includes version 5.14 of KDE Plasma, which can be installed with `apt install kde-standard`.

Plasma has had a rapid evolution based on a very hands-on approach. Its authors quickly got very good results, which allowed them to grow a large user-base. These factors contributed to the overall project quality. Plasma is a mature desktop environment with a wide range of applications.

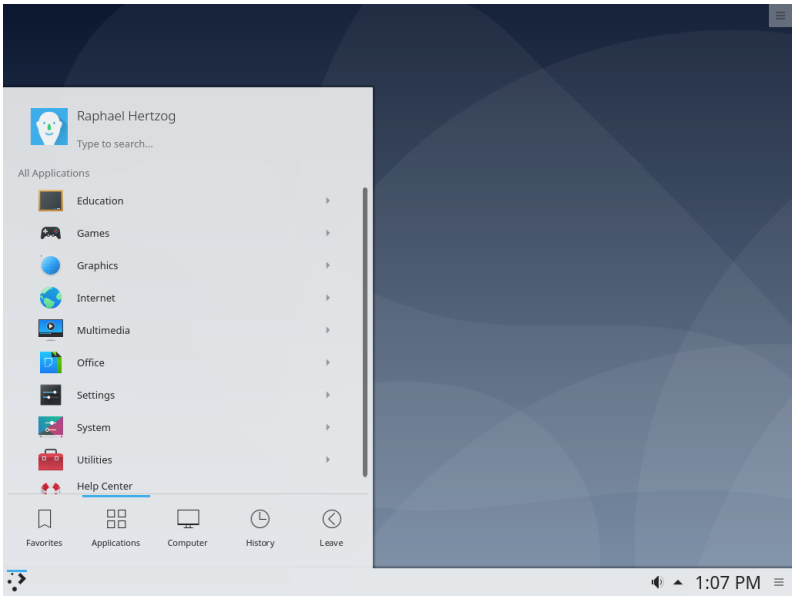


Figure 13.2 *The Plasma desktop*

Since the Qt 4.0 release, the last remaining license problem with KDE software has been solved. This version was released under the GPL both for Linux and Windows (the Windows version was previously released under a non-free license). KDE applications are primarily developed using the C++ language.

13.3.3. Xfce and Others

Xfce is a simple and lightweight graphical desktop, which is a perfect match for computers with limited resources. It can be installed with `apt install xfce4`. Like GNOME, Xfce is based on the GTK+ toolkit, and several components are common across both desktops.

Unlike GNOME and Plasma, Xfce does not aim to become a vast project. Beyond the basic components of a modern desktop (file manager, window manager, session manager, a panel for application launchers and so on), it only provides a few specific applications: a terminal, a calendar (*orage*), an image viewer, a CD/DVD burning tool, a media player (*parole*), sound volume control and a text editor (*mousepad*).

➡ https://xfce.org/

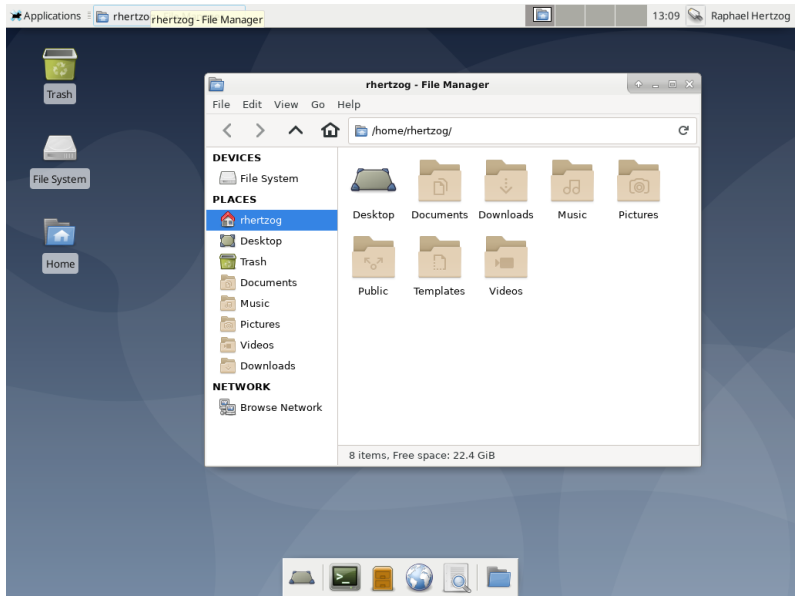


Figure 13.3 *The Xfce desktop*

13.3.4. Other Desktop Environments

LXDE and *LXQt* are two desktop environments focusing on the "lightweight" aspect. The former is GTK+ based while the latter is Qt based. They can be installed with the *lxde* and *lxqt* metapackages.

➨ https://lxde.org/

➨ https://lxqt.org/

Cinnamon and *MATE* both started when GNOME 3 moved away from the traditional desktop paradigm, dropping the usual panel and its menu in favor of the new search-based shell. The former reintroduced a panel by forking GNOME Shell and the latter is a continuation of GNOME 2. They can be installed wih the *cinnamon-desktop-environment* and *mate-desktop-environment* metapackages.

➨ https://developer.linuxmint.com/projects/cinnamon-projects.html

➨ https://mate-desktop.org/

13.4. Email

13.4.1. Evolution

COMMUNITY

Popular **packages**

Installing the *popularity-contest* package enables participation in an automated survey that informs the Debian project about the most popular packages. A script is run weekly by `cron` which sends an anonymized list of the installed packages (by HTTP or email) and the latest access date for the files they contain. This allows the Debian maintainers to know which packages are most frequently installed, and of these, how frequently they are actually used.

This information is a great help to the Debian project. It is used to determine which packages should go on the first installation disks. The installation data is also an important factor used to decide whether to remove a package with very few users from the distribution. We heartily recommend installing the *popularity-contest* package, and participating in the survey.

The collected data are made public every day.

➨ `https://popcon.debian.org/`

These statistics can also help users to choose between two packages that seem otherwise equivalent. Choosing the more popular package is probably a safer choice.

Evolution is the GNOME email client and can be installed with `apt install evolution`. It is more than a simple email client: it also provides a calendar, an address book, a task list, and a memo (free-form note) application. Its email component includes a powerful message indexing system, and allows for the creation of virtual folders based on search queries on all archived messages. In other words, all messages are stored the same way but displayed in a folder-based organization, each folder containing messages that match a set of filtering criteria.

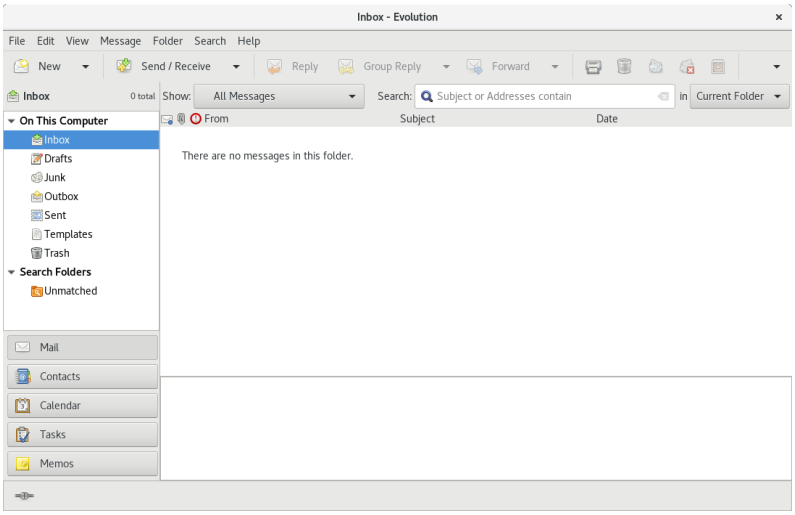


Figure 13.4 *The Evolution email software*

An extension to Evolution allows integration with a Microsoft Exchange email system; the required package is *evolution-ews*[1].

13.4.2. KMail

The KDE email software can be installed with `apt install kmail`. KMail only handles email, but it belongs to a software suite called KDE-PIM (for *Personal Information Manager*) that includes features such as address books, a calendar component, and so on. KMail has all the features one would expect from an excellent email client.

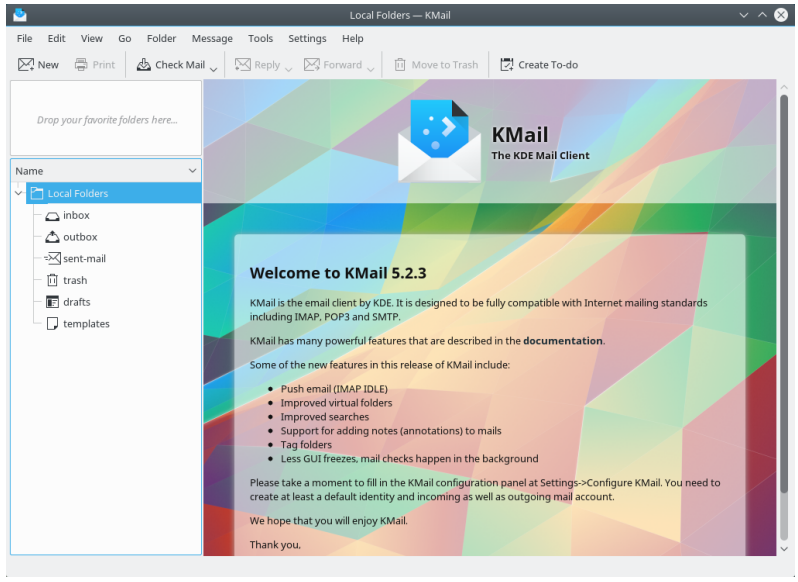


Figure 13.5 *The KMail email software*

13.4.3. Thunderbird

The *thunderbird* package provides the email client from the Mozilla software suite. Various localization sets are available in *thunderbird-l10n-** packages; the *enigmail* extension handles message encrypting and signing, but it is not available in all languages.

[1]The *evolution-ews* package is not part of Debian *Buster*. It was removed during the release process due to a security issue. But at the time of writing a recent version is available as backport (see section 6.1.2.4, "Stable Backports" page 112).

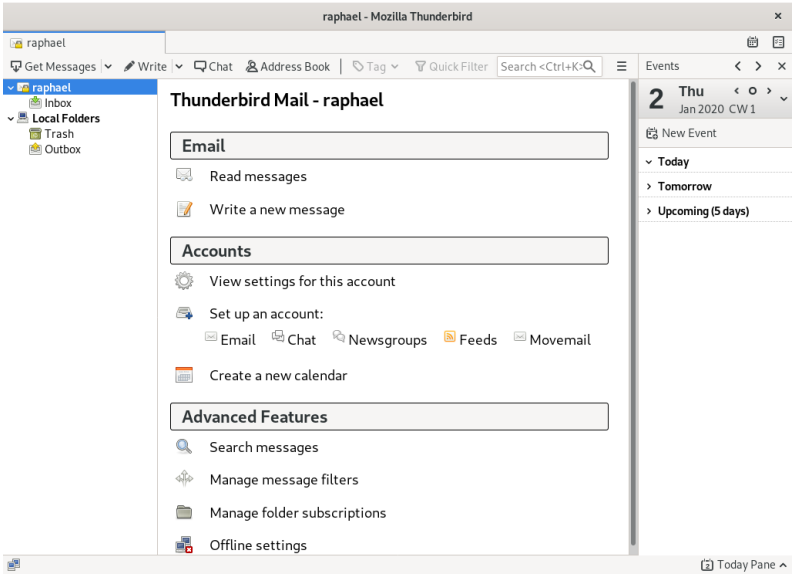


Figure 13.6 *The Thunderbird email software*

13.5. Web Browsers

Epiphany, the web browser in the GNOME suite, uses the WebKit display engine developed by Apple for its Safari browser. The relevant package is *epiphany-browser*.

Konqueror, available in the *konqueror* package, is KDE's web browser (but can also assume the role of a file manager). It uses the KDE-specific KHTML rendering engine; KHTML is an excellent engine, as witnessed by the fact that Apple's WebKit is based on KHTML.

Users not satisfied by either of the above can use Firefox. This browser, available in the *firefox-esr* package, uses the Mozilla project's Gecko renderer, with a thin and extensible interface on top.

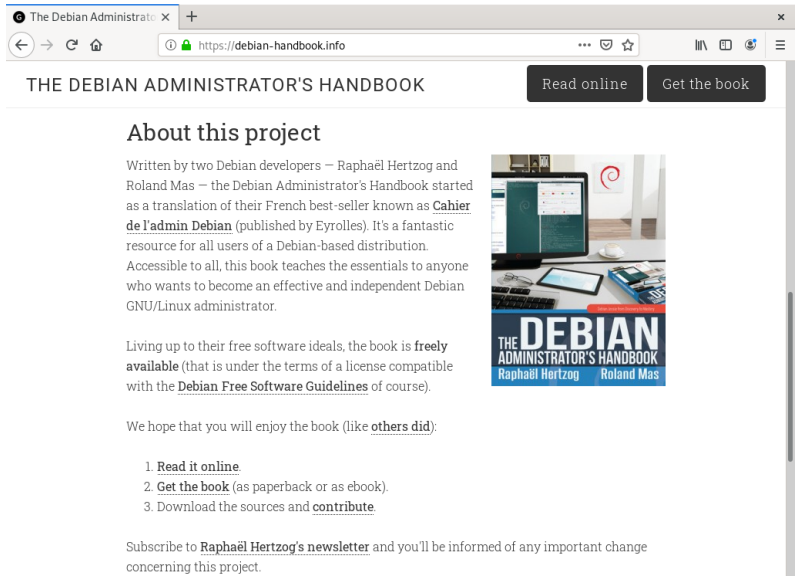


Figure 13.7 *The Firefox web browser*

VOCABULARY

Firefox ESR

Mozilla has a very fast-paced release cycle for Firefox. New releases are published every six to eight weeks and only the latest version is supported for security issues. This doesn't suit all kind of users so, every 10 cycles, they are promoting one of their release to an *Extended Support Release* (ESR) which will get security updates (and no functional changes) during the next 10 cycles (which covers a bit more than a year).

Debian has both versions packaged. The ESR one, in the package *firefox-esr*, is used by default since it is the only version suitable for Debian *Stable* with its long support period (and even there Debian has to upgrade from one ESR release to the next multiple times during a Debian Stable lifecycle). The regular Firefox is available in the *firefox* package but it is only available to users of Debian *Unstable*.

CULTURE

Iceweasel, Firefox and others

Before Debian *Stretch*, Firefox and Thunderbird were missing. The *iceweasel* package contained Iceweasel, which was basically Firefox under another name.

The rationale behind this renaming was a result of the usage rules imposed by the Mozilla Foundation on the Firefox™ registered trademark: any software named Firefox had to use the official Firefox logo and icons. However, since these elements are not released under a free license, Debian could not distribute them in its *main* section. Rather than moving the whole browser to *non-free*, the package maintainer choose to use a different name.

For similar reasons, the Thunderbird™ email client was renamed to Icedove in a similar fashion.

Nowadays, the logo and icons are distributed under a free software license and Mozilla recognized that the changes made by the Debian project are respecting their trademark license so Debian is again able to ship Mozilla's applications under their official name.

CULTURE

Mozilla

Netscape Navigator was the standard browser when the web started reaching the masses, but lost ground when Microsoft bundled Internet Explorer with Windows and signed contracts with computer manufacturers which forbade them from pre-installing Netscape Navigator. Faced with this failure, Netscape (the company) decided to "free" its source code, by releasing it under a free license, to give it a second life. This was the beginning of the Mozilla project. After many years of development, the results are more than satisfying: the Mozilla project brought forth an HTML rendering engine (called Gecko) that is among the most standard-compliant. This rendering engine is in particular used by the Mozilla Firefox browser, which is one of the major browsers.

Last but not least, Debian also contains the *Chromium* web browser (available in the *chromium* package). This browser is developed by Google and has become the most popular browser in just a few years. Its clear purpose is to make web services more attractive, both by optimizing the browser for performance and by increasing the user's security. The free code that powers Chromium is also used by its proprietary version called Google Chrome™.

13.6. Development

13.6.1. Tools for GTK+ on GNOME

Anjuta (in the *anjuta* package) and GNOME Builder (in the *gnome-builder* package) are Integrated Development Environments (IDE) optimized for creating GTK+ applications for GNOME. Glade (in the *glade* package) is an application designed to create GTK+ graphical interfaces for GNOME and save them in an XML file. These XML files can then be loaded by the GTK+ shared library though its GtkBuilder component to recreate the saved interfaces; such a feature can be interesting, for instance for plugins that require dialogs.

➨ `https://wiki.gnome.org/Apps/Builder`

➨ `http://anjuta.org/`

➨ `https://glade.gnome.org/`

13.6.2. Tools for Qt

The equivalent applications for Qt applications are KDevelop by KDE (in the *kdevelop* package) for the development environment, and Qt Designer (in the *qttools5-dev-tools* package) for the design of graphical interfaces for Qt applications.

KDevelop is also a generic IDE and provides plugins for other languages like Python and PHP and different build systems.

13.7. Collaborative Work

13.7.1. Working in Groups: *groupware*

Groupware tools tend to be relatively complex to maintain because they aggregate multiple tools and have requirements that are not always easy to reconcile in the context of an integrated distribution. Thus there is a long list of groupware packages that were once available in Debian but have been dropped for lack of maintainers or incompatibility with other (newer) software in Debian. This has been the case with PHPGroupware, eGroupware, and Kolab.

➨ `https://www.egroupware.org/`

➨ `https://www.kolab.org/`

All is not lost though. Many of the features traditionally provided by "groupware" software are increasingly integrated into "standard" software. This is reducing the requirement for specific, specialized groupware software. On the other hand, this usually requires a specific server. Citadel (in the *citadel-suite* package), Sogo (in the *sogo* package) and Kopano (in the *kopano-core* package) are alternatives that are available in Debian *Buster*.

13.7.2. Collaborative Work With FusionForge

FusionForge is a collaborative development tool with some ancestry in SourceForge, a hosting service for free software projects. It takes the same overall approach based on the standard development model for free software. The software itself has kept evolving after the SourceForge code went proprietary. Its initial authors, VA Software, decided not to release any more free versions. The same happened again when the first fork (GForge) followed the same path. Since various people and organizations have participated in development, the current FusionForge also includes features targeting a more traditional approach to development, as well as projects not purely concerned with software development.

FusionForge can be seen as an amalgamation of several tools dedicated to manage, track and coordinate projects. These tools can be roughly classified into three families:

- *communication*: web forums, mailing-list manager, and announcement system allowing a project to publish news
- *tracking*: tools to track project progress and schedule tasks, to track bugs, feature requests, or any other kind of "ticket", and to run surveys
- *sharing*: documentation manager to provide a single central point for documents related to a project, generic file release manager, dedicated website for each project.

Since FusionForge largely targets development projects, it also integrates many tools such as CVS, Subversion, Git, Bazaar, Darcs, Mercurial and Arch for source control management (also called "configuration management" or "version control"). These programs keep a history of all the revisions of all tracked files (often source code files), with all the changes they go through,

and they can merge modifications when several developers work simultaneously on the same part of a project.

Most of these tools can be accessed or even managed through a web interface, with a fine-grained permission system, and email notifications for some events.

FusionForge is not part of Debian *Stable*. It is a large software stack that is hard to maintain properly and benefits only few users who are usually expert enough to be able to backport the package from Debian *Unstable*.

ALTERNATIVE

GitLab

FusionForge has been used to power the `alioth.debian.org` platform used by the Debian project and its developers for collaborative package management and development for almost a decade. Due to some limitations it has been replaced and shut down in 2018 by a new service powered by GitLab. See sidebar "GitLab, Git repository hosting and much more" page 19.

13.8. Office Suites

Office software has long been seen as lacking in the free software world. Users require replacements for Microsoft tools such as Word and Excel, but these are so complex that replacements were hard to develop. The situation changed when Sun released the StarOffice code under a free license as OpenOffice, a project which later gave birth to LibreOffice, which is available on Debian. The KDE project also has its own office suite, called Calligra Suite (previously KOffice), and GNOME, while never offering a comprehensive office suite, provides AbiWord as a word processor and Gnumeric as a spreadsheet. The various projects each have their strengths. For instance, the Gnumeric spreadsheet is better than OpenOffice.org/LibreOffice in some domains, notably the precision of its calculations. On the word processing front, the LibreOffice suite still leads the way.

Another important feature for users is the ability to import Microsoft Office documents. Even though all office suites have this feature, only the ones in OpenOffice.org and LibreOffice are functional enough for daily use.

THE BROADER VIEW

LibreOffice replaces OpenOffice.org

OpenOffice.org contributors set up a foundation (*The Document Foundation*) to foster the project's development. The idea had been discussed for some time, but the actual trigger was Oracle's acquisition of Sun. The new ownership made the future of OpenOffice under Oracle uncertain. Since Oracle declined to join the foundation, the developers had to give up on the OpenOffice.org name. This office suite is now known as *LibreOffice*, and is available in Debian.

After a period of relative stagnation on OpenOffice.org, Oracle donated the code and associated rights to the Apache Software Foundation, and OpenOffice is now an Apache project. This project is not currently available in Debian and is rather moribund when compared to LibreOffice.

LibreOffice and Calligra Suite are available in the *libreoffice* and *calligra* Debian packages, respectively. Although the *gnome-office* package was previously used to install a collection of office